## Introduction

Outlier detection is a diagnostic step, not a cleaning step: flagged points should be investigated, not automatically removed. Several rules produce flags from different premises; each has a different trade-off between sensitivity and robustness. This page covers the common univariate rules and their assumptions.

## Prerequisites

The reader should know what mean, median, SD, and IQR are.

## Theory

- **1.5 x IQR fence (Tukey)**: observations outside $[Q_1 - 1.5 \cdot \text{IQR},\ Q_3 + 1.5 \cdot \text{IQR}]$ are mild outliers; $3 \cdot \text{IQR}$ marks extremes. Robust and distribution-free.
- **Z-score (3-sigma)**: $|z_i| > 3$ where $z_i = (x_i - \bar{x})/s$. Sensitive to the outliers it is trying to flag (masking).
- **Hampel identifier**: $|x_i - \text{median}| > 3 \cdot \text{MAD}$. Robust variant of the z-score that uses median and MAD.
- **Grubbs' test**: formal test of the most extreme value under normality. Gives a p-value but is sensitive to its own normality assumption.

Masking and swamping are two pitfalls: masking occurs when true outliers inflate the SD and hide themselves; swamping occurs when one outlier drags neighbours across the threshold, falsely flagging them.

## Assumptions

- IQR fence and Hampel: no distributional assumption; robust.
- Z-score: approximately normal distribution.
- Grubbs' test: normality and a single suspected outlier.

## R Implementation

```r
library(outliers)

set.seed(2026)
x <- c(rnorm(48, mean = 50, sd = 8), 95, 110)

# IQR fence
q <- quantile(x, c(0.25, 0.75))
iqr <- IQR(x)
fences <- c(q[1] - 1.5 * iqr, q[2] + 1.5 * iqr)
which(x < fences[1] | x > fences[2])

# Z-score
z <- (x - mean(x)) / sd(x)
which(abs(z) > 3)

# Hampel
h <- (x - median(x)) / mad(x)
which(abs(h) > 3)

# Grubbs' test for the single most extreme
grubbs.test(x)
```

## Output & Results

```
IQR fence flags:   [49, 50]
z-score flags:     [50]
Hampel flags:      [49, 50]
Grubbs: G = 2.81, p = 0.03
```

The IQR and Hampel rules identify both extreme values; the z-score, inflated by those same values, identifies only the most extreme. Grubbs' test rejects the null that the extreme point is from the same distribution.

## Interpretation

Report which rule you applied and how many observations it flagged. "Two observations exceeded the Tukey 1.5 x IQR upper fence; both were verified as accurate data-entry values and retained in the primary analysis" is a defensible sentence.

## Practical Tips

- Flag, investigate, decide. Never delete silently.
- For skewed data, IQR fences still flag normal-range observations as outliers; consider log-transformation first.
- For multivariate outliers, use Mahalanobis distance; for regression diagnostics, use Cook's distance.
- Grubbs' test assumes normality; for non-normal data prefer the IQR or Hampel rule.
- Report sensitivity analyses that re-run the primary analysis with and without flagged observations; the conclusion should be robust.

## For Reviewers

**What to look for in a paper using this method.**

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

## See also — labs in this chapter

- Descriptive statistics and Table 1